Plan the electric car

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use circuit and mechanical evidence to plan a controlled motor-car test.

SUCCESS CRITERIA - I can...

1. Draw the battery-switch-motor circuit.
2. Predict effects of an open switch and a changed connection.
3. Plan one controlled performance comparison.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A magnet moves through a 50-turn coil. Propose two controlled changes that could increase induced-emf magnitude and explain each.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: P = IV; E = PΔt.
Constant power model. Use approved battery, switch and motor. Disconnect before rewiring; keep wheels clear for first power.

Worked reasoning: A classroom car circuit uses a battery, switch and motor in one closed loop. The switch opens the current path; a chassis drawing must also reserve space for safe wiring and moving parts.

Your turn: Draw the circuit separately from a labeled component-placement plan.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Draw and label the battery, switch, motor and wires as a complete loop. Mark a location where an open connection stops current.



Circuit symbols: long/short lines = battery; rectangles = resistors; M = motor.

Marzano check - circle: ① ② ③

04 Apply the physics

A supplied model-car trial covers 4 m in 4 s. Calculate average speed. Then prepare to record three trials for your own car. Leave distance and time entries blank until measured.

Trial	Distance (m)	Time (s)	Average speed (m/s)
1	____	____	____
2	____	____	____
3	____	____	____

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Your team changes wheels, battery and axle alignment together and gets a faster run. Can you attribute the change to wheels alone? Explain a better comparison.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A current of 4 A flows for 6 s. Find the charge that passes. Draw the stated circuit and label quantities relevant to the question.

Draw the complete circuit and label shared nodes.

Marzano check - circle: 1 2 3

07 Exit check

List one safety check and one measurement you need before the first powered test.



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Design two switch placements that both interrupt the same series motor circuit. Explain one practical reason to choose between them.



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3